

OPTICAL INSTRUMENTS

Optical Instruments

Optical instruments are devices that use lenses or mirrors to form and magnify images. Important instruments include the simple microscope, compound microscope and astronomical telescope.

Human Eye

The human eye is a natural optical instrument. The eye lens forms a real and inverted image on the retina. The iris controls the amount of light entering the eye and the pupil is the opening through which light enters.

Power of Accommodation

The ability of the eye lens to change its focal length so that objects at different distances can be focused clearly on the retina is called accommodation.

Near Point and Far Point

For a normal human eye, the near point is about 25 cm and the far point is at infinity. The least distance of distinct vision is denoted by D .

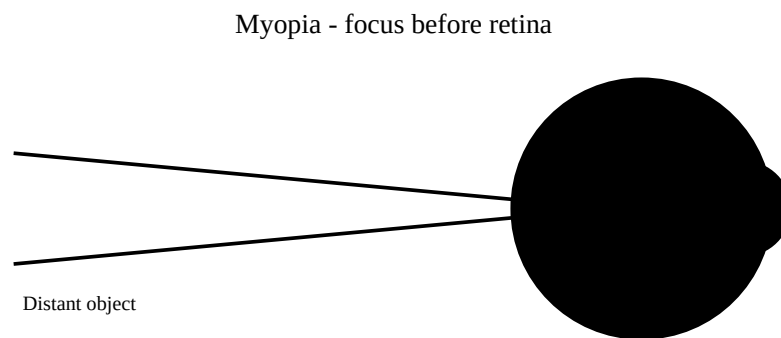
Defects of Vision

The important defects of vision are myopia, hypermetropia and presbyopia. These defects occur when the eye cannot form a sharp image on the retina.

Myopia

In myopia, a distant object is focused in front of the retina. A concave lens is used to correct myopia.

Myopia — Correction

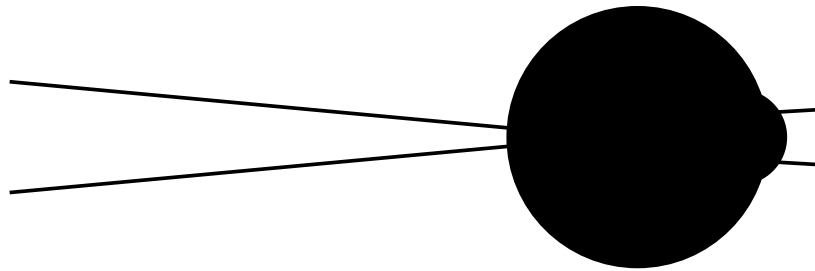


Hypermetropia

In hypermetropia, a nearby object is focused behind the retina. A convex lens is used to correct hypermetropia.

Hypermetropia — Correction

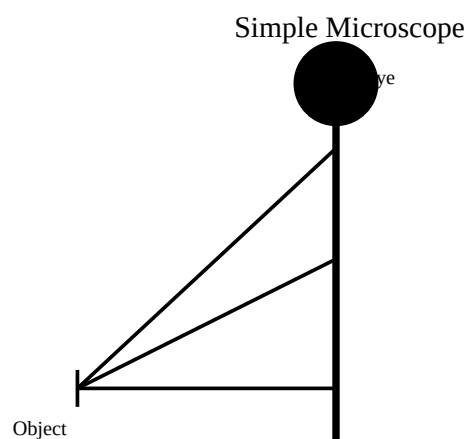
Hypermetropia - focus behind retina



Correction of Vision Defects

Myopia is corrected using a concave lens, while hypermetropia is corrected using a convex lens. Proper lens power is selected according to the defect.

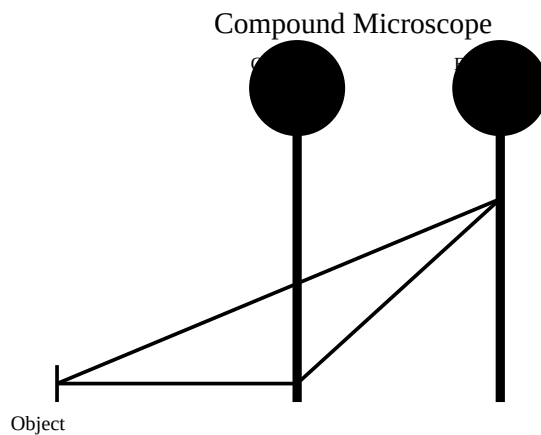
Simple Microscope



Magnifying Power of Simple Microscope

For normal adjustment, magnifying power $M = D / f$. When the final image is formed at the least distance of distinct vision, $M = 1 + D / f$.

Compound Microscope

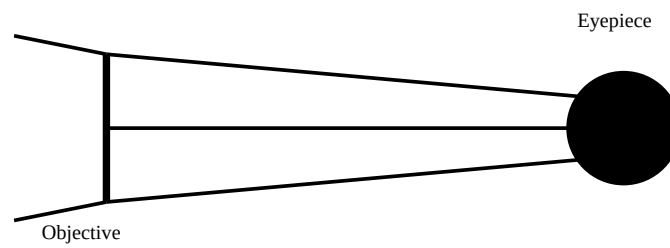


Magnifying Power of Compound Microscope

For normal adjustment, magnifying power $M = (L / f_o) (D / f_e)$, where L is the tube length, f_o is the focal length of the objective and f_e is the focal length of the eyepiece.

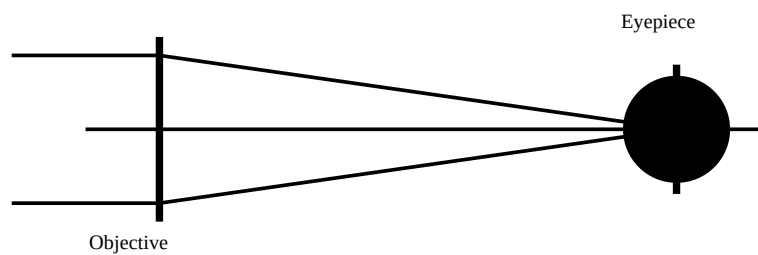
Astronomical Telescope

Astronomical Telescope



Telescope — Normal Adjustment

Telescope - Normal Adjustment



Magnifying Power of Telescope

For an astronomical telescope in normal adjustment, magnifying power $M = f_o / f_e$. The image is inverted.

Ray Diagram of Microscope

In a compound microscope, the objective first forms a real, inverted and magnified intermediate image. The eyepiece acts as a simple microscope and produces the final magnified virtual image.

Ray Diagram of Telescope

The objective of a telescope forms a real image of a distant object near its focal plane. The eyepiece magnifies this image for the observer.

Least distance of distinct vision: $D = 25 \text{ cm}$

Simple microscope: $M = D / f$

Simple microscope at near point: $M = 1 + D / f$

Compound microscope: $M = (L / f_o) (D / f_e)$

Telescope in normal adjustment: $M = f_o / f_e$

Length of telescope in normal adjustment: $L = f_o + f_e$

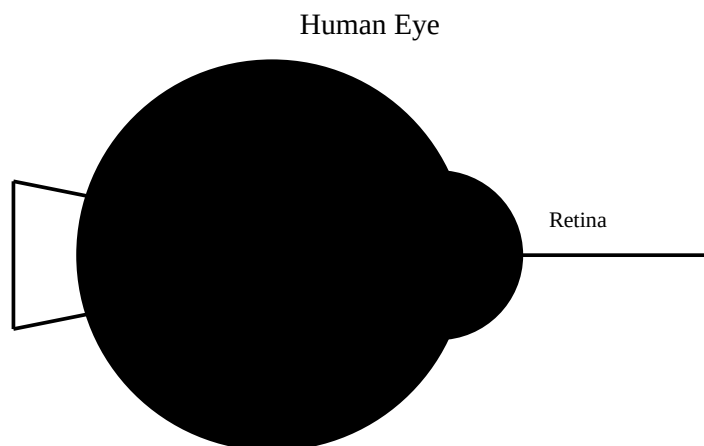
Quick Revision Points

Myopia — concave lens. Hypermetropia — convex lens. Simple microscope — one convex lens. Compound microscope — objective + eyepiece. Astronomical telescope — large focal length objective and small focal length eyepiece. Normal adjustment means final image at infinity.

MHT-CET Important Points / Numerical Focus

Focus on lens power, focal length, magnifying power, least distance of distinct vision, tube length, objective focal length and eyepiece focal length. Remember sign conventions and units while solving numerical problems.

Human Eye — Diagram



Magnifying Power — Diagram

Magnifying Power

